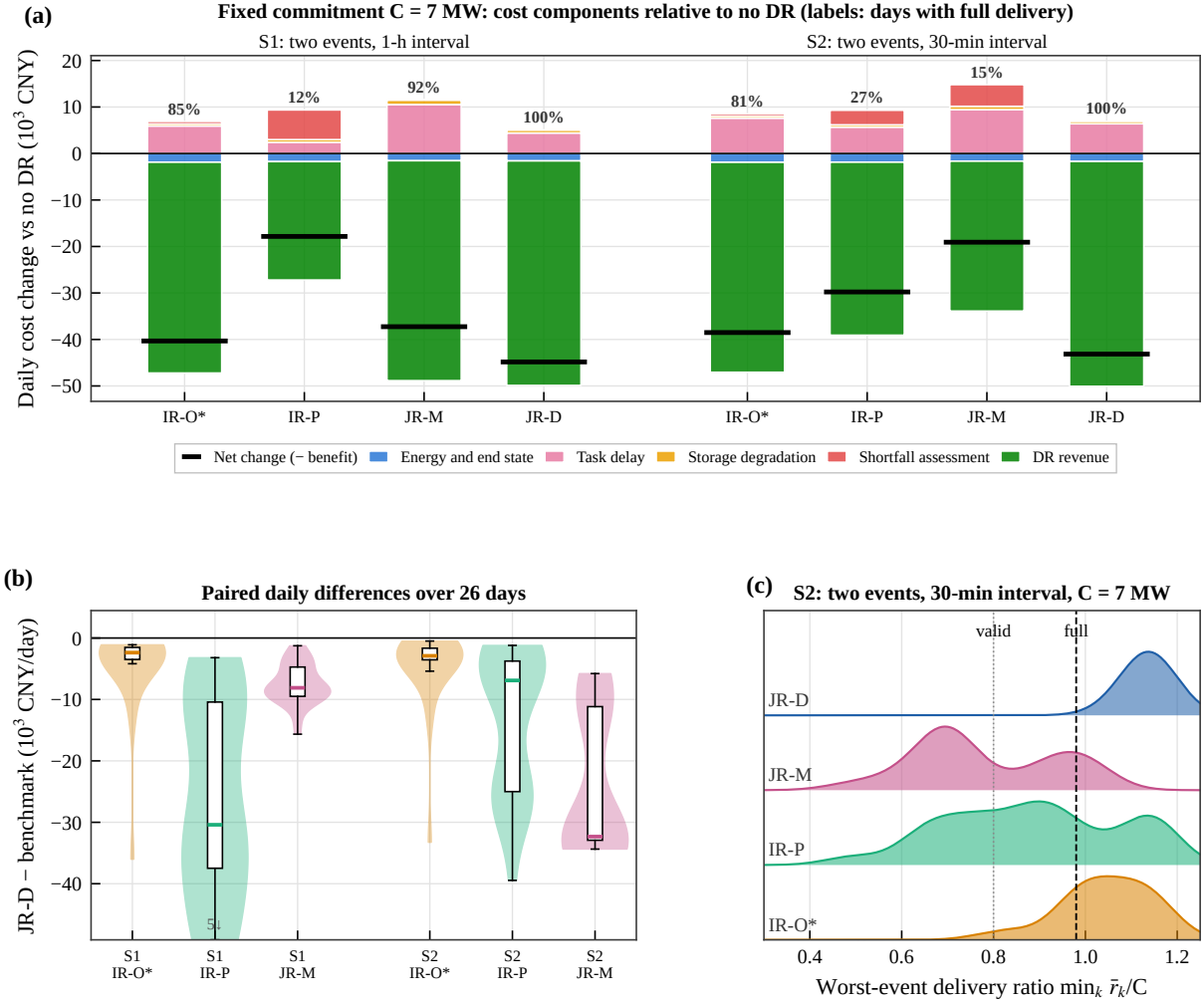

(a)
Fixed commitment C = 7 MW: cost components relative to no DR (labels: days with full delivery)
S1: two events, 1-h interval
S2: two events, 30-min interval
Daily cost change vs no DR (10^3 CNY)
20
10
0
−10
−20
−30
−40
−50
85%
12%
92%
100%
81%
27%
15%
100%
IR-O*
IR-P
JR-M
JR-D
IR-O*
IR-P
JR-M
JR-D
Net change (− benefit)
Energy and end state
Task delay
Storage degradation
Shortfall assessment
DR revenue
(b)
Paired daily differences over 26 days
JR-D − benchmark (10^3 CNY/day)
0
−10
−20
−30
−40
54
S1 IR-O*
S1 IR-P
S1 JR-M
S2 IR-O*
S2 IR-P
S2 JR-M
(c)
S2: two events, 30-min interval, C = 7 MW
valid
full
JR-D
JR-M
IR-P
IR-O*
0.4
0.6
0.8
1.0
1.2
Worst-event delivery ratio $\min_k \bar{r}_k/C$